

Drone Teleoperation Interfaces: Challenges and Opportunities with XR Integration

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Abstract

Drones are piloted remotely through teleoperation interfaces. These interfaces typically employ 2D displays, which can limit Situation Awareness (SA) due to narrow field-of-view (FoV) and lack of depth perception. Extended Reality (XR) has potential to enhance these systems by providing 3D environments that improve spatial understanding and expand FoV. However, designing drone teleoperation interfaces requires a thorough understanding of remote pilot's SA needs and usability challenges to ensure safety and facilitate decision-making. This makes a user-centered design (UCD) approach essential. To this end, we present findings from qualitative interviews with (n=8) professional drone pilots, designed to (1) identify key challenges in current teleoperation interfaces, and based on these challenges, (2) provide an outlook on how XR could address these issues. This work aims to guide future research and design of XR-based teleoperation interfaces to improve SA and safety in drone operations.

CCS Concepts

• **Human-centered computing** → **User centered design**; *Mixed / augmented reality*.

Keywords

drone teleoperation challenges, UAV, remote piloting, situation awareness, UCD, extended reality, XR

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1 Introduction

Unmanned Aerial Vehicles (UAV), commonly known as drones, are expected to transform our civil airspace with operations ranging from search and rescue missions [7], to cargo and passenger transport [25]. Consequently, Human-Drone Interaction (HDI) is growing as a research area within the Human-Computer Interaction (HCI) community [2, 38]. Drones are piloted remotely, which creates a loss of sensory cues that would otherwise provide the pilot with visual information from an out-of-the-window view [16]. As such, visual feedback of the remote environment is an important feature of the teleoperation interface. Processed sensor data from the drone are presented through teleoperation interfaces, which serve as the interaction point of the remote operator with the drone. The presented data usually includes flight instrument data, a map to represent the drone's position, and, if available, video feed from the cameras mounted in the drone, displayed typically on 2D screens [36]. These setups, although functional, present significant limitations in Situation Awareness (SA) [12, 22]. SA represents remote operator's capacity to (Level 1) observe, (Level 2) comprehend, and (Level 3) project future events and conditions within the UAV's operational surroundings [9]. High SA is essential for remote operator's decision-making. Building and maintaining SA in drone teleoperation is challenging due to information overload presented on limited Field-of-View (FoV) screens, which lack peripheral vision and depth perception [36]. Poor depth cues reduce remote operator's ability to determine the exact 3D distance between the drone and the potential surrounding obstacles [18, 19].

These challenges highlight the need for better design of teleoperation interfaces. That being said, Extended Reality (XR) technologies, encompassing the spectrum from Augmented Reality (AR) to Virtual Reality (VR) [28, 34], are being considered across various safety-critical domains as a solution for increased SA [42]. XR-based interfaces can more accurately visualize a 3D world, which in turn may facilitate the comprehension of spatial relationships, distances, and terrain features [36], increasing spatial awareness. Furthermore, they support multi-modal interaction, which can reduce cognitive load, enhance usability, and improve overall user satisfaction [17]. Despite the promising potential of XR in UAV operations, research in this area remains relatively underexplored, with existing work

focusing on prototyping initial solutions [1, 35] that do not deliberately design for improved SA, as recommended in the literature [10].

Designing drone teleoperation systems with emerging technologies like XR necessitates adopting a User-Centered Design (UCD) approach. This proactive involvement helps prevent design mistakes that could impose potential risks later on [17]. To this end, we conducted qualitative interviews with $n = 8$ expert remote operators, including professional pilots and researchers to understand challenges with existing teleoperation interfaces and explore the potential of XR technologies in addressing these challenges. Thus, our contribution is twofold, (1) we provide a categorization of teleoperation challenges based on qualitative insights from domain experts, and (2) we draw initial recommendations on how XR technologies can address these challenges.

2 Related Work

Drone teleoperation challenges. Drones can range across various sizes and automation levels, from no automation to full autonomy, enabling hobbyist and professional applications. In this paper, we address professional drone operations, typically involving larger drones, where human-in-the-loop control remains essential to ensure safety and task completion [18]. Safety is the primary concern in professional drone operations [26], and challenges in teleoperation can significantly compromise it. Laubner et al. [22] identified the main challenges in remote piloting, namely, fault tolerance and reliability of the system, communication latency, and situational awareness. Hobbs et al. [16] focused on human factor challenges. They included unconventional characteristics of drone control, such as monotonous ultra-long flights, lack of sensory information, control transfer, reliance on automation, as well as physical characteristics of the control station as some of the main challenges. Furthermore, in [15], authors identified link quality, control transfer, mode error, maps, weather, and other challenges as causes of critical incidents during remote piloting. A recent study [21] revisited human factor challenges in unmanned aviation and emphasized the need for more research due to fast technological advancements in the field. Our research focuses on identifying challenges in teleoperation interface design that contribute to poor situational awareness.

XR in drone teleoperation interfaces. Several studies highlight the potential of XR technologies to improve situational awareness and reduce cognitive load in drone teleoperation [17, 22, 26, 27]. Authors in [22] suggest that XR interfaces could address SA issues by expanding remote operators' FoV and enhancing performance. Although XR technologies have already been implemented in drone teleoperation, a large number of existing works are proof-of-concept. They primarily demonstrate the feasibility of using XR technologies in drone teleoperation, including mission planning [30, 39] and mission execution tasks, e.g., for navigation [43], landing support [1], and target detection in reconnaissance procedures [35]. Some of these studies present performance and usability evaluations through user studies. However, direct SA evaluations (e.g. using SAGAT or SART) [11] remain limited, with most studies relying on performance measures only, such as task completion time or error rate, which are insufficient to evaluate SA [11]. While XR-based teleoperation interfaces have shown promising results, it

is not clear what features, associated to XR interfaces, enable the most effective means for operators to, for instance, view, search, or detect relevant objects in the virtual space, which would be key to a better SA and lower cognitive load. This raises the question: How should XR-based teleoperation interfaces be designed to support SA in UAV operations?

Designing for situation awareness. Endsley et al. [10] provide guidance for creating SA-oriented system designs by taking a UCD approach. The framework comprises three key steps (1) SA Requirements Analysis with domain experts, and Technology Analysis to inform strategic decisions on technology selection; (2) SA Design, which integrates established human factors guidelines for interface development; and (3) SA Evaluation, ensuring that the system effectively supports the operator's SA and overall performance. In this work, we focus on step 1.

Few previous studies have taken a UCD approach to explore requirements for teleoperation interface design in specific drone use cases, e.g., in future urban air mobility [27], and search and rescue operations [14]. Similarly, studies in related domains, such as autonomous driving [13, 37], have applied UCD methods and involved domain experts early in the design process. However, these studies have not yet considered the potential of XR technologies. Building on these works, our work takes a different perspective. We aim to identify and categorize general challenges related to existing teleoperation interfaces that span across different drone use cases, enabling broader applicability. We also discuss the potential of XR technologies before designing future prototypes. This ensures that future design process is guided by user requirements and the capabilities of emerging technologies, offering valuable insights for researchers and designers developing XR-based teleoperation interfaces.

3 Method

To identify the key challenges in drone teleoperation interfaces, we conducted in-depth semi-structured interviews with domain experts including remote pilots and researchers in the field working with drone teleoperation systems. While focus groups were considered, individual interviews were chosen to capture the diverse experiences of each remote operator and allow more dedicated time for each participant. The small sample size at this step allows us to prioritize thick and thorough data [8]. Below we describe the process in more detail.

Participants. We interviewed eight domain experts, aged 29 to 49 ($M=37$, $SD=6.16$). All of them were researchers in the unmanned aviation domain, working closely with teleoperation interfaces in various drone operational tasks. Six of them were also professional drone pilots, conducting experimental and research flights in different mission types, including humanitarian aid over long distances. This dual expertise gives them a forward-looking perspective on challenges and opportunities in drone teleoperation, informed by both practical operations and research-driven innovation. The interviewed experts had from 6 to 20 years of experience with drone teleoperation interfaces. We aimed for a balanced representation of genders, however, all eight participants identified as male. Informed consent and confidentiality procedures were followed. An overview of all participants is given in Table 1.

Table 1: Overview of study participants

	Gender	Age	Occupation	Experience with teleoperation	Experience with XR
P1	M	29	Drone Pilot, Researcher	9-10 years	3-4 years
P2	M	40	Researcher	8 years	No
P3	M	37	Drone Pilot, Researcher	6 years	No
P4	M	31	Drone Pilot, Researcher	7 years	2 years
P5	M	39	Drone Pilot, Researcher	11 years	No
P6	M	34	Researcher	6-7 years	No
P7	M	37	Drone Pilot, Researcher	6-7 years	No
P8	M	49	Drone Pilot, Researcher	15-20 years	2-3 years

Data collection and analysis. The interviews were conducted during March - July 2024, face-to-face, with one exception, conducted through video-conferencing. The interviews lasted between 45 minutes and 1 hour. We began each interview by outlining its purpose and obtaining the participant’s consent for audio recording. We followed the same structure and set of questions for each interview and adapted to the discussions with the participants. The main questions asked during the interview were: (1) What conditions make planning and/or flying a drone difficult? Can you share an example of a challenging scenario you encountered during drone teleoperation? What was the biggest challenge there? (2) What aspects of the user interface (UI) are limiting your work mostly? Are there any changes that would make your work easier? (3) What information do you need to build SA? What information do you need to have on the display at all times? We also asked them about their experience with XR, and their outlook on the benefits and limitations of XR in drone operations.

The interviews were audio recorded, transcribed automatically, and subsequently reviewed and corrected by the interviewer. Data analysis followed an inductive approach following the guidelines of thematic analysis from Braun and Clarke [4]. An initial coding round was conducted by the first author. Subsequently, a second coding round, independent of the first one, was done by the second author. Once the codes were established, they were iteratively analyzed to identify key themes which will be presented in Section 4.

4 Results

In the analysis of the interviews, several key challenges were identified and coded. The main themes that emerged from the data include: spatial awareness, user interface, environmental conditions, and coordination and collaboration. Each theme entails subcategories of challenges remote operators experience, as seen in Fig 1. The following sections will delve into each theme, presenting specific insights derived from the interviews. This will be followed by the operator’s outlook on the integration of XR technologies in drone teleoperation in section 4.5.

4.1 Spatial Awareness

Situation awareness encompasses several kinds of awareness, e.g. system awareness (battery, other sensors), mode awareness (e.g. manual, automated), task awareness (mission-specific steps), and spatial awareness, which is the understanding of spatial relationships in the 3D environment, including the position of the aircraft

in this environment [40, 41]. This aspect of SA is critical when designing drone teleoperation interfaces, precisely because pilots are not sitting in the aircraft and have limited sensory information, i.e they lack an out-of-window view.

Sensory input limitations. Remote pilots rely on teleoperation interfaces displaying telemetry data and video feeds from on-board sensors and First Person View (FPV) cameras. Nevertheless, the limited FoV restricts operators’ peripheral vision and spatial awareness. These challenges are more pronounced in Beyond Visual Line-of-Sight (BVLOS) flights, where pilots have no direct visual contact with the drone. P3 commented that flying BVLOS feels like “a computer game,” requiring trust in the system due to the lack of direct sensory cues. This description can also be found in [29]. In contrast, Visual Line-of-Sight (VLOS) flights allow operators to maintain direct visual contact with the aircraft. However, as P3 highlighted: “*..the main problem with VLOS flights is that you usually look at the drone. So, if you want to check the battery or check some other data, you always have to have this head-down moment...*”. Reduced sensory cues limit the operator’s spatial awareness, which is also in line with previous literature findings [16, 22].

Map accuracy. All participants mentioned the importance of maps for planning and executing drone operations but raised concerns about missing obstacles, such as buildings, trees, or power lines. They mentioned this often necessitates on-site visits to manually identify and measure these obstacles. However, some pilots emphasized that going on-site to map the environment is not always feasible, or even possible in cases where missions require ad-hoc planning, as P1 mentioned: “*..I have to go to the real site, have a look ..which is fine for smaller missions, but not really for actual missions, ..e.g. applied in humanitarian aid, where you have ad-hoc planning and you can’t just go and look at every bit of the mission first...*”. This shows that the current approaches are limited and bring safety challenges, especially during remote emergency missions. This finding also correlates with existing literature [15, 27, 32].

Position, distance, and scale estimation. Spatial mapping in 2D brings cognitive challenges when it comes to understanding spatial relationships (distances, positions, scale), primarily due to limitations in the field-of-view and in depth cues [41]. Remote operators emphasized the difficulty in accurately estimating drone’s position and scale in relation to the surrounding environment. P2 pointed out “*I think the most challenging is to estimate your real position in the world and the distance to other parts of the world, and the scale of what you look at...*”. This is critical especially during landing, as also noted in [27].

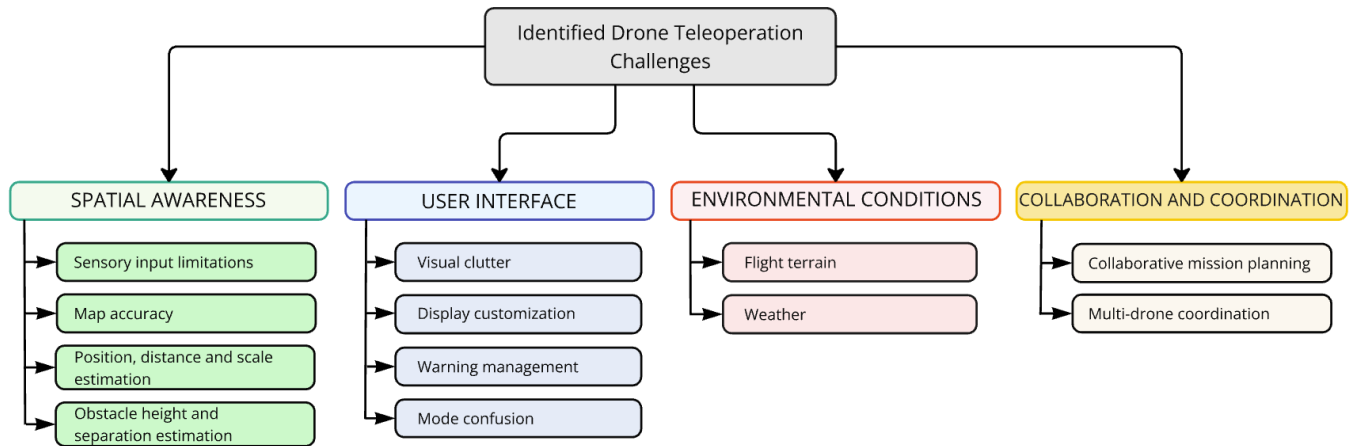


Figure 1: Categorization of challenges with drone teleoperation interfaces

Obstacle height and separation estimation. Drone altitude data from sensors is presented numerically on the teleoperation interface, but there is no numerical data for obstacle height. Pilots must mentally estimate safe separation from obstacles, which can lead to uncertainty, especially in high obstacle areas, where maintaining safe separation levels is crucial. As P3 explained: “Sometimes I would like to have the altitude of the drone more visible... so you have a 2D map and you see the position, that’s fine. You can also see the speed, but the altitude you can’t see from the picture. You always have to read from digits. And then you have to create [mental image].”

4.2 User Interface

Visual clutter. Remote operators often feel overwhelmed by the excessive information displayed on the teleoperation interface. In addition to primary mission tasks, operators must monitor battery levels, sensor warnings, and other air traffic. P1 shared: “..so I would say one thing [challenge] is the amount of information you get. You can focus on only a small sub-sample of the information you get...you can look at everything but you have to decide basically what is most relevant”. P8 also illustrated: “..when dealing with turbulence or unexpected issues, you naturally narrow your focus to the most important points and then you can’t keep track of everything..”. This finding is in line with previous literature [12, 16].

Display customization. Operators mentioned the lack of customization in teleoperation interfaces as a challenge, describing how this restricts their ability to optimize the interface for different operational needs. As P5 explained: “..I think you should have different views, one for normal operation, where you have just limited data because everything is working as expected. This way, you can focus on the mission itself. He further explained: “..there are times when you want to understand why the drone is behaving a certain way..”, and mentioned that in those cases “..you need all the data to really debug that behavior..”. Furthermore, P8 highlighted that some information can be hidden in different sub-windows and that you have to actively select it to view it, which can create delays in critical moments.

Warning management. Effective warning management is crucial to ensure operators can quickly identify and respond to urgent issues. However, operators revealed problems which include poorly visible warning colors, rapid disappearance of alerts, and a lack of clarity on the relevance of specific warnings. P1 described: “..for example, ..you have warnings pop-up but you can’t actually get the warning before it’s gone again, and you don’t know what was actually this warning. So that’s not so ideal. Or you get a lot of warnings and you don’t know what is now relevant..”. In addition, remote operators noted that color-coded warnings can be difficult to see in bright sunlight if they are outside in VLOS operations, which affects their visibility and, consequently, operators’ ability to respond quickly. This finding is also corroborated by previous literature [12].

Mode confusion. Depending upon their mode of operation capabilities, UAVs can be operated manually, partially autonomous, and fully autonomous [5]. As these systems grow in complexity and automation level, the risk of mode confusion increases. This is a phenomenon known in aviation where the operators believe they are in a mode different than the one they are actually in [20], which can compromise safety. P8 described: “..if you have a system with open-source components, which can have 10 or 15 different modes with no standardized user interface, you need to find out which mode to select to achieve the result you want. And it is challenging to understand what is the drone doing, we call it mode confusion..”. This finding is also supported by the existing literature [15].

4.3 Environmental Conditions

Flight terrain. All participants mentioned that high obstacle environments, especially mountains and urban areas with high buildings, bring challenges to SA more than flat environments. Mountains also impede mobility, so flying BVLOS is often the solution. Considering UAV flights are limited in the altitude they can and are allowed to reach, remote pilots described that when flying over high mountains there is a risk of flying too low, creating a small margin for error. P2 shared their experience on an earthquake crisis management scenario: “..So that was a challenging terrain, flying as low as possible to get good camera images, and at the same time

not getting too close to mountains in a hilly area. The operator further described *".. we tried to move the camera and change the flight attitude, and try to guess what the distances are to the ground and to the mountains."* Likewise, remote pilots mentioned urban environments present challenges that come from a high presence of obstacles and safety considerations that populated areas bring. P2 noted: *"..everything in an urban environment that has obstacles [is challenging], ..some database or sensor information would be very relevant.."*. Whereas P6 highlighted: *"The most difficult missions are ..flying in the city because of the safety reasons. The drone has to have a parachute system or something special, or to be not very heavy, in case of a crash.."*.

Weather. Weather is a critical factor in aviation, and even more critical for lighter aircraft such as drones. One of the major ways in which wind for example affects UAVs is by changing its flight trajectory, which can result in deviations from planned paths and increased risk of accidents [33]. Pilots frequently highlighted the challenges they face in wind conditions and expressed a need for integrated real-time weather updates and visualizations of wind direction. P5 illustrated: *"..a challenging environment is always when the drone does not follow the path as it should ..because of wind gusts, which could be even worse in urban scenarios.. So it would be interesting to see the wind directions.. to have a feeling where will my drone glide to when I have to eject the parachute, or where is it dangerous if we are drifting off the planned path because of wind"*. Other adverse weather conditions were also mentioned like rain, and extreme temperatures, but drone flights in such weather conditions are still avoided. This finding is also present in previous literature [14, 21, 33].

4.4 Collaboration and Coordination

Collaborative mission planning. Remote operators reported to be co-located when planning missions collaboratively. In situations where co-location is not possible, they mentioned relying on traditional video-conferencing tools. However, they expressed the need for better collaboration tools that enable all team members to be aware of the same spatial perspectives during co-planning. P8 tells more from their experience: *"..If you could optimize this collaboration so that the persons don't need to be on the same place, that would be easy. And then you need to get the same impressions [perspective] of the challenge that you want to tackle, ..having the same view on the problem, .. same zoom level for example"*. This calls for further research in providing better remote collaborative mission planning tools, as also found in [32].

Multi-drone coordination. Managing multiple drones in the same airspace introduces additional safety challenges. Operators stated that effective coordination and communication are essential to avoid collisions and ensure safe operations. P7 shared their insights: *"Having multiple drones simultaneously airborne in the same area is challenging. Coordination with different drone pilots, often from various organizations, becomes tricky. More than three drones in a small area requires careful communication and pre-flight planning. Contingency and emergency procedures are crucial in case something goes wrong"*. This highlights important future research directions in the control of drone swarms as well as in unmanned air traffic management, already addressed in the existing literature [14, 24].

4.5 Remote Operators' Outlook on XR Integration

Seven out of eight participants expressed a strong interest in XR technologies, particularly for improving situational awareness. P1, referring to BVLOS flights, stated: *"..what I really would like to have is a VR or XR headset replacing my 2D limited view.."*. He further suggested that synthetic vision in XR headsets could provide an expanded FoV alongside camera views, enabling a more accurate judgement of drone's position, e.g. during landing. P7 and P3 noted the benefits of AR in VLOS operations. P3 stated: *"If I have this [telemetry data] in AR, like a head-up display, I don't need a second person, really. Because the questions I would ask them, I could check myself.."*. For mission planning, pilots envisioned VR with 3D representations of terrain and airspace boundaries to *"..really see it from all angles.."*, as P5 highlighted. Whereas P7 noted: *"I think VR could be used in mission planning, e.g. to see is my flight route avoiding all obstacles.."*.

P2 expressed an overall skepticism about XR: *"I couldn't imagine that I feel better when looking at some XR display as opposed to a good 2D view.."*. The rest of the participants, despite seeing potential, expressed their concerns about XR's usability. A common concern was that the prolonged use of XR headsets can be tiring. P4 mentioned *"..at least for an hour it's okay, but in two hours it gets quite warm and it's tiring.."*. Additionally, P7 expressed concerns for potential distractions: *"I see the fear that AR/VR could distract the drone pilot from actually monitoring/ controlling the drone. Too much is also not helpful.."*.

5 Discussion

As UAV operations expand into high-obstacle environments, i.e., mountains and dense urban environments, often requiring BVLOS flights, the need for new teleoperation display paradigms becomes increasingly evident. This study has empirically explored challenges remote operators face with current drone teleoperation interfaces, and has assessed their opinion on using XR technologies as a solution to these challenges. The findings, consistent with existing literature [15–17, 21, 22], highlight limitations on several aspects of SA, namely *spatial awareness, user interface, environmental factors*, and, *collaboration and coordination*. In addition, remote pilots' outlook on future XR-based interfaces was positive, as indicated in related literature [3].

XR technologies present a promising solution to improve *spatial awareness*, by providing peripheral vision through a wide FoV [18, 36]. Likewise, they offer practical solutions to *user interface* challenges by enabling customizable, multi-modal interaction through hand gestures, voice commands, and touch controls [17]. Moreover, synergy with AI can be leveraged to develop adaptive context-aware interfaces [31]. Additionally, XR can improve *collaboration* in remote mission planning by providing synchronized 3D views, reducing reliance on conventional video-conferencing tools. XR can also facilitate the *coordination of multiple drones* [36] by integrating sensor data, such as thermal imaging, LiDAR, and GPS, from all involved drones into a unified operational view, supporting control and decision-making.

Initial recommendations that could help address some of the identified challenges are outlined below.

- For BVLOS operations develop VR/MR interfaces that combine synthetic vision with on-board camera feeds to provide a full view of the environment. Enable seamless perspective switching between first-person view, third-person view, and side-views to enhance situation awareness and control.
- For VLOS flights, AR as a Head-Up Display (HUD) is recommended. Keep important telemetry data (e.g., altitude, speed, battery status), persistently visible within the operator's FoV to minimize head-down time.
- Develop trajectory guidance tools that visualize predicted flight paths, especially for manual control in critical situations (e.g., indicating where the drone will be if a specific turn angle is maintained).
- Provide tools that visualize safe separation distances (e.g. transparent tunnel) to assist in collision avoidance with obstacles.
- Augment terrain awareness features (e.g., grid-map) to facilitate safe navigation and landing in complex environments.
- Integrate external data sources into the interface, e.g., real-time weather updates, especially wind direction, flight volumes, and population density maps to provide a complete operational picture.

Overall, implementing these technologies requires addressing practical constraints such as fatigue when used longer [6], cyber-sickness [23], e.g., when VR/MR is used in BVLOS flights, and outdoor visibility when AR HUD displays are used in VLOS flights. We assess that at this stage of technological development, XR can complement existing teleoperation tools rather than replace them entirely, allowing operators to use XR selectively, e.g., in mission validation, and in complex missions. While this study provides valuable insights into the challenges of UAV teleoperation, its limitations should be acknowledged. The small sample size ($n=8$) may limit generalizability, although does not necessarily decrease validity [8]. Moreover, the study relied on qualitative data, which may be subject to participant bias. Although we contextualized their insights, the findings should be interpreted with caution.

6 Conclusion

We provided insights into drone pilot's challenges in the context of teleoperation interfaces and the potential of XR technologies to assist them. A semi-structured interview with eight professional remote pilots revealed that future drone teleoperation requires improved situational awareness through advanced tools and visualizations that enhance spatial understanding, especially in high-obstacle environments, as well as customizable user interfaces and improved collaborative tools. Different XR technologies could be leveraged to address these challenges for specific flight types, e.g. AR HUDs for VLOS flights and MR/VR for BVLOS flights. With the continuous advancement of UAV systems, it is important to consider the implications of new forms of display paradigms. Future research should expand on these findings by designing and testing XR prototypes in real-world drone operations. Careful consideration of task-specific needs and usability constraints is essential here to realize the full potential of XR in this domain.

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A Interview Questions

Table 2: Interview Questions

Demographics and Experience Context	
1	What is your professional occupation?
2	For how long have you been working with drone teleoperation interfaces? How often do you work with them?
3	What teleoperation system/software do you use? Can you describe your most common tasks there?
4	Are you familiar with XR technology? Do you have experience with it? If yes, how much?
What constitutes a challenging operation	
5	What environmental conditions make planning and/or flying difficult?
6	There are different missions/operations types that can be done with drones. Which ones are more challenging to plan, and/or fly through the current interfaces that you use?
7	Can you share an example of a challenging scenario encountered during mission planning and/or mission execution? What was the biggest challenge?
Limitations of current teleoperation UIs	
8	In your experience and opinion, what aspects of the user interface (UI) are limiting your work?
Collaboration	
9	How you collaborate and communicate with other colleagues or involved parties?
Information needs for situation awareness	
10	What information do you need to always have on the screen to be able to do your work?
11	What are the first data points in the display that you typically look at to build situational awareness?
12	Are there cases when you need to change the perspective from the birds eye view to the 3D view?
a.	How often do you use 3D view? Why? Do you think XR would help here?
Exploring future XR solutions	
13	Based on your experience, what interactions or visualizations in future XR interfaces would better support your work?
14	What limitations do you foresee when integrating XR with drones?